



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23									
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (Minors)									
(Applicable for CE, ECE, EEE&ME)									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Year/ Sem	Cr	L	T	P	C.I.E	S.E.E	Total Marks
B23ADM101	Fundamentals of Artificial Intelligence	II-II	3	3	0	0	30	70	100
B23ADM201	Fundamentals of Data Science	III-I	3	3	0	0	30	70	100
B23ADM301	Cloud Computing	III-II	3	3	0	0	30	70	100
B23ADM401	Introduction to Prompt Engineering & Gen AI	IV-I	3	3	0	0	30	70	100
B23ADM501	*MOOCS-I	II-II to IV-I	3	--	--	--	--	--	100
B23ADM601	*MOOCS-II	II-II to IV-I	3	--	--	--	--	--	100
TOTAL			18	12	0	0	120	280	600

*Two MOOCS courses of any ARTIFICIAL INTELLIGENCE AND DATA SCIENCE related Program Core Courses from NPTEL/SWAYAM with a minimum duration of 12 weeks (3 Credits) courses other than the courses offered need to be taken by prior information to the concern. These courses should be completed between II Year II Semester to IV Year I Semester.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADM101	Minors	3	--	--	3	30	70	3 Hrs.
FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE								
(Minor Degree course in AIDS)								
Course Objectives:								
1.	To have a basic proficiency in a traditional AI language including an ability to write simple to intermediate programs and an ability to understand code written in that language							
2.	To understand the basic issues of knowledge representation and blind and heuristic search, as well as an understanding of other topics such as minimax, resolution that play an important role in AI programs							
3.	To have a basic understanding of some of the more advanced topics of AI							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1	Describe the basic foundations & applications of AI.							K2
2	Apply problem solving strategies to generate the best AI solutions using state space search.							K3
3	Use AI languages to represent knowledge base for problem solving.							K3
4	Use AI tools to represent knowledge base in real world problems.							K3
5	Apply uncertainty and fuzzy logic techniques to solve AI problems							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction, history, intelligent systems, foundations of AI, applications, tic-tac-toe game playing, development of AI languages, current trends.							
UNIT-II (10 Hrs)	Problem solving: state-space search and control strategies: Introduction, general problem solving, characteristics of problem, exhaustive searches, heuristic search techniques, iterative deepening A*, constraint satisfaction Problem reduction and game playing: Introduction, problem reduction, game playing, alpha beta pruning, two-player perfect information games.							
UNIT-III (10 Hrs)	Logic concepts: Introduction, propositional calculus, propositional logic, natural deduction system, axiomatic system, semantic tableau system in propositional logic, resolution refutation in propositional logic, predicate logic.							
UNIT-IV (8 Hrs)	Knowledge representation: Introduction, approaches to knowledge representation, knowledge representation using semantic networks, extended semantic networks for KR, knowledge representation using frames Advanced knowledge representation techniques: Introduction, conceptual dependency theory, script structure							

UNIT-V (12Hrs)	Expert system and applications: Introduction phases in building expert systems, expert system versus traditional systems Uncertainty measure: probability theory: Introduction, probability theory, Bayesian belief networks Fuzzy sets and fuzzy logic: Introduction, fuzzy sets, fuzzy set operations, types of membership functions
Textbooks:	
1.	Artificial Intelligence- Saroj Kaushik, CENGAGE Learning.
2.	Artificial intelligence, A modern Approach, 2nded, Stuart Russel, Peter Norvig, PEA.
Reference Books:	
1.	Artificial Intelligence- Deepak Khemani, TMH, 2013.
2.	Introduction to Artificial Intelligence, Patterson, PHI.
3.	Artificial intelligence, structures and Strategies for Complex problem solving, George F Luger, 5th ed, PEA.



Course Code:B23ADM101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
II B.Tech. II Semester MODEL QUESTION PAPER					
FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE					
(Minor Degree course in AIDS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Mention any two applications of AI.	1	2	2
	b).	What is an intelligent system? Give an example.	1	2	2
	c).	What is alpha-beta pruning?	2	2	2
	d).	What is the difference between breadth-first and depth-first search?	2	2	2
	e).	Define resolution refutation.	3	2	2
	f).	What is propositional logic?	3	2	2
	g).	What is a conceptual dependency?	4	2	2
	h).	Mention the use of frames in knowledge representation.	4	2	2
	i).	List any two phases of building expert systems.	5	2	2
	j).	What is the key difference between traditional systems and expert systems?	5	2	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	List various categorizations of artificial intelligence systems. Explain each.	1	2	5
	b).	Explain various fields in foundations of AI.	1	2	5
		OR			
3.	a).	Describe the working of a simple AI game – Tic-Tac-Toe.	1	3	5
	b).	What are the applications of Artificial Intelligence?	1	2	5
		UNIT-2			
4.	a).	Describe Iterative Deepening A* algorithm.	2	3	5
	b).	Describe the mini max algorithm with an example	2	3	5
		OR			
5.	a).	Explain Constraint Satisfaction Problem (CSP) and solve a Cryptarithmic puzzle (TWO+TWO=FOUR), show the steps involved in finding the solution.	2	3	5
	b).	What is meant by search strategy? Explain any two search strategies	2	2	5

		that come under uniformed search			
		UNIT-3			
6.		Consider the following problem. • John likes all kinds of food. • Apples are food. • Chicken is food. • Anything any one eats and isn't killed by is food. • Bill ate peanuts and still alive. • Sue eats everything Bill eats. i) Convert the formulas into clause form. ii) Prove that "John likes peanuts" using resolution.	3	3	10
		OR			
7.	a).	Prove the following theorem using deductive inference rules From $A \rightarrow B \wedge C$, A infer C , from $A \wedge B$, $A \rightarrow C$ infer C .	3	3	5
	b).	What is predicate logic? Explain the predicate logic representation with reference to suitable example	3	3	5
		UNIT-4			
8.	a).	Define frames. Explain knowledge representation using frames?	4	2	5
	b).	Describe your chair using a semantic net?	4	3	5
		OR			
9.	a).	Develop a frame based system for university application?	4	2	5
	b).	How do you represent visiting a restaurant in the form of a Script? Explain	4	3	5
		UNIT-5			
10.	a).	What is Expert system? Explain its Phases.	5	2	5
	b).	Explain Fuzzy sets with example.	5	2	5
		OR			
11.	a).	Explain various types of membership functions used in fuzzy logic with diagrams.	5	2	5
	b).	Discuss Bayesian belief networks with an example.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADM201	Minors	3	--	--	3	30	70	3 Hrs.
FUNDAMENTALS OF DATA SCIENCE								
(Minor Degree course in AIDS)								
Course Objectives:								
1	Impart knowledge on basics of data science, data manipulation and exploratory data analysis concepts that are vital for data science.							
2	Develop skills for applying tools and techniques to analyze, visualize and interpret data.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1	Demonstrate knowledge on the concepts of data science to perform mathematical computations using efficient storage and data handling methods in NumPy.							K2
2	Apply Data Preparation and Exploration methods using Pandas to perform data manipulation.							K3
3	Apply Data Cleaning and preparation of data.							K3
4	Apply data visualization techniques using Matplotlib and Seaborn							K3
5	Apply methods to analyze and interpret time series data to extract meaningful statistics.							K3
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION TO DATA SCIENCE: Basic terminologies of data science, Types of data, five steps of data science, Arrays and vectorized computation using NumPy - The NumPy ndarray: A multidimensional array object, Universal functions: Fast element-wise Array functions, Array-oriented Programming with arrays, File input and output with arrays, Linear algebra, Pseudo random number generation.							
UNIT-II (10 Hrs)	DATA EXPLORATION WITH PANDAS: Process of exploring data, Panda's data structures – Series, Data frame, Index objects; Essential functionality, Summarizing and computing descriptive statistics – Correlation and covariance, Unique values, Value counts and membership; Data loading, Storage, and file formats - Reading and writing data in text format.							
UNIT-III (10 Hrs)	DATA CLEANING, PREPARATION AND DATA WRANGLING: Handling missing data, Data transformation, String manipulation - String object methods, Data wrangling: Hierarchical indexing, Combining and merging datasets: join, Combine and reshape and pivoting.							

UNIT-IV (10 Hrs)	DATA VISUALIZATION WITH MATPLOTLIB: Plotting and visualization- A brief matplotlib API primer, Plotting with Pandas and Seaborn, Other python visualization tools, Apply: General split-apply-combine, Data aggregation and Group operations Group By mechanics.
UNIT-V (8 Hrs)	TIME SERIES ANALYSIS: Date and time data types and tools, Time series basics. Date ranges, Frequencies, Re-sampling and frequency Conversion: Down sampling, up sampling and interpolation, Re-sampling with periods.
Textbooks:	
1.	Wes McKinney, Python for Data Analysis, O 'Reilly, 2nd Edition, 2017
Reference Books:	
1.	Sinan Ozdemir, Principles of Data Science, Packt Publishers, 2nd Edition, 2018.
2.	Rachel Schutt, Cathy O'Neil, Doing Data Science: Straight Talk from the Frontline, O'Reilly, 2014.
e-Resources	
1	https://swayam.gov.in/nd1_noc19_cs60/preview
2	https://towardsdatascience.com/
3	https://www.w3schools.com/datascience/
4	https://github.com/jakevdp/PythonDataScienceHandbook
5	https://www.kaggle.com

Estd. 1980

AUTONOMOUS

Course Code:B23ADM201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
FUNDAMENTALS OF DATA SCIENCE					
(Minor Degree course in AIDS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Define arrange,array,reshape functions	1	2	2
	b).	Create N-dimn Array	1	3	2
	c).	Explain pandas data structures	2	2	2
	d).	Define stack ,unstuck functions	2	2	2
	e).	Describe merge methods	3	2	2
	f).	Define Hierarchical indexing	3	2	2
	g).	Explain data visualization tools	4	2	2
	h).	Difference between pivot, cross tabulation	4	2	2
	i).	Types of moving window functions	5	2	2
	j).	Describe resample.	5	2	2
Estd. 1980					
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	Explain Linear algebra with Numpy	1	2	5
	b).	Build in detail about Data Science process with necessary examples	1	2	5
		OR			
3.	a).	Explain Required steps of data science	1	2	5
	b).	Explain Arrays and vectorized computation using NumPy with example.	1	2	5
		UNIT-2			
4.	a).	How many types of files in data science? Explain each with examples .	2	2	5
	b).	Explain Correlation and covariance with examples.	2	2	5
		OR			
5.	a).	Explain Data loading, Storage using pandas.	2	2	5
	b).	Make a pandas Data Frame with two-dimensional list using python.	2	3	5
		UNIT-3			

6.		Is regular expressions important for data science? What are the applications of regular expression?	3	2	10
		OR			
7.	a).	Explain about Data Wrangling and uses of data wrangling.	3	2	5
	b).	Apply a methods join, Combine and reshape - Hierarchical indexing using student sample data.	3	3	5
		UNIT-4			
8.	a).	Define Data Visualization what are the benefits of data visualization	4	2	5
	b).	Explain Data aggregation and Group operations Group By mechanics.	4	2	5
		OR			
9.	a).	Apply Different types of plots with examples.	4	3	5
	b).	How can we visualize more than three dimensions of data in a single chart?	4	2	5
		UNIT-5			
10.	a).	What is resampling and describe the methods of Down sampling, up sampling with examples.	5	2	5
	b).	Describe the various tools used to represent the time data types	5	2	5
		OR			
11.	a).	Describe various applications of time series data and list out the basics of time series data	5	2	5
	b).	Explain the various methods for Moving window functions	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADM301	Minors	3	--	--	3	30	70	3 Hrs.
CLOUD COMPUTING								
(Minor Degree course in AIDS)								
Course Objectives:								
1	Examine the system models for cloud computing.							
2	Understand the concepts of virtualization, hardware and storage							
3	Identify cloud platform architecture and programming.							
4	Develop cloud applications							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1	Define, understand, and explain the concepts of cloud computing environment and various Virtualization techniques.							K2
2	Explore and understand various services provided by Cloud Computing							K2
3	Illustrate various Cloud application development frame							K2
4	Understand various cloud maintenance techniques							K2
5	Develop cloud-based applications by applying Amazon, Microsoft, Salesforce.com etc., frameworks.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to cloud computing: Cloud computing components, Infrastructure services, storage applications, database services – introduction to SaaS, PaaS, IaaS, IdaaS, data storage in cloud. Virtualization: enabling technologies, types of virtualizations, server virtualization, desktop virtualization, memory virtualization, application and storage virtualization-tools and products available for virtualization.							
UNIT-II (10 Hrs)	SAAS and PAAS: Getting started with SaaS, SaaS solutions, SOA, PaaS and benefits. IaaS and Cloud Data Storage: understanding IaaS, improving performance for load balancing, server types within IaaS, utilizing cloud-based NAS devices, cloud based data storage, backup services, cloud-based block storage and database services.							
UNIT-III (10 Hrs)	Cloud Application development: Client server distributed architecture for cloud designing cloud-based solutions, coding cloud-based applications, traditional Apps vs cloud Apps, client-side programming, server-side programming overview fundamental treatment of web application frameworks.							
UNIT-IV	Cloud Governance and economics: Securing the cloud, disaster recovery and business							

(10 Hrs)	continuity in the cloud, Managing the cloud, migrating to the cloud, governing and evaluating the clouds business impact and economics.
UNIT-V (8 Hrs)	Inside Cloud: Introduction to MapReduce and Hadoop-overview of big data and its impact on cloud. Google: Google App Engine, Google Web Toolkit Microsoft: Azure Services Platform, Windows live, Exchange Online, Share Point Services
Textbooks:	
1.	Cloud Computing: SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and More, Kris Jamsa, Jones & Bartlett Publishers, Paperback edition, 2013
2.	Cloud Computing, A Practical Approach, 1st Edition, Anthony T Velte, Toby J Velte, Robert Elsenpeter, TMH, 2017
Reference Books:	
1.	Hadoop MapReduce cookbook, Srinath Perera and Thilina Gunarathne, Packet publishing.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc21_cs15/preview



Course Code:B23ADM301					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
CLOUD COMPUTING					
(Minor Degree course in AIDS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Compare traditional applications and cloud-based applications.	1	1	2
	b).	What are the benefits of using PaaS for application development?	1	1	2
	c).	What is virtualization? Explain the different types of virtualizations used in cloud computing.	2	1	2
	d).	List commonly used tools and products for virtualization.	2	1	2
	e).	Write a short note on cloud-based block storage.	3	1	2
	f).	Explain the concept of Infrastructure as a Service (IaaS) with a real-world cloud provider example.	3	1	2
	g).	Explain client-side programming vs server-side programming with examples.	4	1	2
	h).	How do cloud-based solutions differ from traditional on-premises systems?	4	1	2
	i).	Explain the MapReduce programming model.	5	1	2
	j).	Discuss the impact of big data on cloud infrastructure and services.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	What is Cloud Computing? Explain Cloud Components with neat diagrams.	1	2	5
	b).	Distinguish Full Virtualization and Para Virtualization.	1	2	5
		OR			
3.	a).	Apply and Distinguish Full Virtualization and Para Virtualization in high-performance computing applications.	1	2	5
	b).	Illustrate the concept of logical clocks and message delivery rules with neat diagram	1	2	5
		UNIT-2			
4.	a).	Analyze open SaaS solutions and mashups with real-time examples. Apply Service-Oriented Architecture diagrams in developing cloud	2	2	5

		application.			
	b).	Build a PaaS application using Google App Engine and Force.com?	2	2	5
		OR			
5.	a).	Apply IaaS concepts for improving performance through Load Balancing with neat diagrams.	2	2	5
	b).	Analyze different types of servers in IaaS. Apply IaaS solution concepts to build on Rackspace.	2	2	5
		UNIT-3			
6.	a).	Discuss Client-Server Distributed Architecture in the Cloud.	3	2	5
	b).	Distinguish between Traditional Apps and Cloud Apps.	3	2	5
		OR			
7.	a).	Apply design principles to build cloud-based solutions.	3	2	5
	b).	Design a web application framework using.	3	2	5
		UNIT-4			
8.	a).	Write an analysis document on Business Continuity and Disaster Recovery in the cloud.	4	2	5
	b).	Explain cloud management strategies.	4	2	5
		OR			
9.	a).	Analyze the concept of Data Storage Wiping for preventing data access.	4	2	5
	b).	Discuss cloud migration strategies.	4	2	5
		UNIT-5			
10.	a).	Write an analysis document on MapReduce and Hadoop.	5	2	5
	b).	Analyze and assess Big Data and its impact on the Cloud.	5	2	5
		OR			
11.	a).	Summarize features of the Google Web Toolkit	5	2	5
	b).	Elaborate on SharePoint services and Exchange Online.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADM401	Minors	3	-	-	3	30	70	3 Hrs.

INTRODUCTION TO PROMPT ENGINEERING & GEN AI

(Minor degree course in AIDS)

Course Objectives: This course is designed to

1.	Understand the principles and techniques of prompt engineering, including the designs of effective prompts
2.	Explore the capabilities of large language models for text and image generations and to leverage the creation of engaging content
3.	Gain practical experience in crafting prompts and generating text and images using AI tools and platforms

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	Design clear, concise, and relevant prompts following the standard principles of prompt engineering	K4
2.	Utilize LLMs to generate text and image for designing more effective content and design	K4
3.	Analyse existing prompts and make strategic combinations for enhanced prompts	K3
4.	Design AI powered application	K4

SYLLABUS

UNIT-I (10Hrs)	Introduction to Large Language Models: What are Text Generation Models, LLMs in the Market. Understanding Prompting and Prompt Techniques: Five Principles of Prompting, Importance of prompt, prompt engineering process, setting objectives, prompt design, Evaluation of responses, prompt refining, prompt templating, Challenges and Limitations of Using Prompts.
UNIT-II (8 Hrs)	The Art of Text Data Generation with GenAI Standard Practices for Text Generation with ChatGPT: Basics of Generating Lists, Explain It Like I'm Five, Universal Translation Through LLMs, Ask for Context, Text Style Unbundling, Identifying the Desired Textual Features, Generating New Content with the Extracted Features, Role Prompting.
UNIT-III (10 Hrs)	Generating Text with LangChain: Introduction to LangChain, Chat models, Streaming chat models, Creating Multiple LLM Generations, LangChain prompt templates, Using prompt template with chat models,

	creating Few -shot prompt templates, Limitations with few shot examples.
UNIT-IV (10 Hrs)	Introduction to Diffusion Models for Image Generation: Introduction to Image Generation with AI, Available Models - OpenAI DALL-E, Stable Diffusion, Google Gemini, Text to Video, Model Comparison. Standard Practices for image generation with Midjourney: Reverse Engineering Prompts, Negative Prompts, Prompt Re-Writing, Prompt Analysis.
UNIT-V (8 Hrs)	Building AI Powered Applications: AI Blog Writing, Topic Research, Expert Interview, Generate Outline, Text Generation, Writing Style, Title Optimization, AI Blog Images, User Interface.
Textbooks:	
1.	James Phoenix, Mike Taylor, “Prompt Engineering for Generative AI”, O’Reilly, To Release in May 2024 https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/
2.	Gilbert Mizrahi, “Unlocking the Secrets of Prompt Engineering: Master the Art of Creative Language Generation to Accelerate Your Journey from Novice to Pro”, January 2024 https://www.packtpub.com/en-in/product/unlocking-the-secrets-of-prompt-engineering9781835083833
Reference Books:	
1.	Michael Ferguson, “Prompt Engineering: The Future of Language Generation”, January 2023 https://books.apple.com/us/book/prompt-engineering-the-future-of-language-generation/id6445529200
2.	“Prompt Engineering Guide”, https://www.promptingguide.ai/
3.	“Prompt Engineering for Generative AI”, Google, https://developers.google.com/machinelearning/resources/prompt-eng

Course Code: B23ADM401					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO PROMPT ENGINEERING & GEN AI					
(Minor Degree course in AIDS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Large Language Model (LLM). Give two examples of popular LLMs used in Generative AI.	1	2	2
	b).	List any two principles of effective prompting.	1	2	2
	c).	What is “Explain It Like I’m Five (ELI5)” prompting?	2	1	2
	d).	What is role prompting? Give one example.	2	2	2
	e).	What is LangChain? Mention one advantage of using LangChain in prompt engineering.	3	2	2
	f).	What is a prompt template in LangChain?	3	2	2
	g).	What is a diffusion model in image generation?	4	2	2
	h).	What is a negative prompt in image generation?	4	2	2
	i).	What is title optimization in AI blog writing?	5	2	2
	j).	List any two components of an AI-powered blog writing workflow.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the concept of Large Language Models (LLMs) and discuss the major LLMs available in the market.	1	2	5
	b).	Describe the prompt engineering process with suitable example, including objective setting, prompt design, evaluation, and refinement.	1	2	5
		OR			
3.	a).	Explain the five principles of prompting with suitable examples.	1	2	5
	b).	Discuss the challenges and limitations of using prompts in Generative AI systems.	1	2	5
		UNIT-2			
4.	a).	Explain standard practices for text generation using ChatGPT with suitable examples.	2	2	5
	b).	Discuss the importance of textual feature extraction in generating high-quality AI content.	2	2	5
		OR			
5.	a).	Explain the concept of universal translation through LLMs with an	2	2	5

		example.			
	b).	Describe role prompting and text style unbundling with suitable examples.	2	2	5
		UNIT-3			
6.	a).	Explain the architecture and working of LangChain in text generation applications.	3	2	5
	b).	Discuss the use of prompt templates in LangChain with suitable examples.	3	2	5
		OR			
7.	a).	Explain few-shot prompt templates in LangChain with an example.	3	2	5
	b).	Discuss the limitations of few-shot prompting in LangChain applications.	3	2	5
		UNIT-4			
8.	a).	Explain diffusion models and their role in AI-based image generation.	4	2	5
	b).	Compare DALL·E, Stable Diffusion, and Google Gemini for image generation.	4	2	5
		OR			
9.	a).	Explain reverse engineering prompts in Midjourney with suitable examples.	4	2	5
	b).	Discuss negative prompts and prompt rewriting in image generation systems.	4	2	5
		UNIT-5			
10	a).	Explain the workflow for building an AI-powered blog writing application.	5	2	5
	b).	Discuss the role of topic research, expert interview, and outline generation in AI blog writing.	5	2	5
		OR			
11.	a).	Explain how AI can be used for title optimization and writing style enhancement in blog generation.	5	2	5
	b).	Discuss the importance of AI-generated images and user interface design in AI-powered applications.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks